

Zian Fang

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EDUCATION

University of Wisconsin - Madison | Madison, WI August 2020 – May 2024
Bachelor of Science, Double Major in Computer Science and Economics Cumulative GPA: 3.57/4.00
Certificate: Game Design, Japanese (Fluent)
Courses: Software Engineering, Intro-Artificial Intelligence, Operating Systems, Theory of Computing

SKILLS

Programming Languages: C, C++, C#, Java, Python, Rust, HTML/CSS/JavaScript, MATLAB, STATA, R, Assembly
Software: Postman, Perforce, Git, MySQL, AWS, Unity, Unreal Engine, Expo
Frameworks and Operating Systems: React, Node.js, PyTorch, TensorFlow, Linux

INTERNSHIP EXPERIENCE

Shanghai Lilith Technology Co. Shanghai, China
Machine Learning Engineer, Tools C++, Rust September 2024 – Present

- Participated in the development of an AI-native game built in Unreal Engine 5; developed the locomotion system, character and animation blueprints, and interfaces in game bot behaviors.
- Re-engineered behavior trees for game Farlight 84, extending imitation learning from combat to non-combat behaviors such as looting, enhancing bot efficiency and realism.

Shanghai Lilith Technology Co. Shanghai, China
Machine Learning Engineer, Tools: PyTorch, C++, Rust June 2023 – September 2023

- Built and trained a double-layer image-classification model for the anti-cheat system in the game Farlight84 based on mobilenetv3 (client-side) and efficientnetv2 (server-side), identifying 6,000+ cheating accounts daily with a mis-judgement rate below 0.01% based on validation set, safeguarding the integrity of game ecosystem.
- Developed an AI-based story generator for *Call of Dragon* using MiniGPT-4 to transform uploaded images into the game's art style and generate a story, receiving 90%+ positive feedback; Self-deployed and fine-tuned the AI model, resulting in significant cost savings while maintaining comparable output quality.
- Participated in the early development of an AI-generated game by developing a text-to-speech model using VITS and pretraining an in-house LLM, and supported personalization across plot, lore, dialogue and character behavior/animations.

PROJECT EXPERIENCE

AI Assistant for Course Scheduling (Python) | *Software Engineering* | Madison, WI June 2024 – Present

- Worked with an education tech startup on automating complex course scheduling tasks for non tech savvy K-12 teachers.
- Developed a text-to-course-schedule API using GPT-4, allowing teachers to create schedules through natural language input.
- Built an Python-based API to streamline user requests, integrate current schedules, and generate final course plans, significantly saving time for course planning and improving efficiency.

SQL-Based File System (Python, JavaScript, HTML, CSS) | *Software Engineering* | Madison, WI January 2024 – May 2024

- Built a file system with an authentication mechanism, allowing users to create, edit, move and delete files and directories as a team.
- Automate backend testing by creating test scripts and deploying them in the CI/CD pipeline, ensuring testing of every code push and preventing build-on of undetected bugs; Implemented the user-friendly frontend interface for visualizing files and directories.
- Managed product backlogs, user stories, user requirements and test plans, coordinating with teammates on timelines.

AI Agent (JavaScript, HTML, CSS) | *Front-End Developer* | Madison, WI January 2024 – May 2024

- Implemented and trained a chat-style AI agent with WitAI to interpret user input and generate corresponding responses
- Developed a web application using React, enabling real-time interactions;
- Built an app version using Expo, extending the accessibility of the AI agent across mobile platforms.

Torchwood Game (C#) | *Programmer* | Madison, WI September 2023 – December 2023

- Built a turn-based pixel-style 2D RPG game, including farming, crafting and fighting with monsters
- Implemented Unity for game development and collaborated with teammates on game flow and level design